

Midterm Assignment: Prompt Engineering Course

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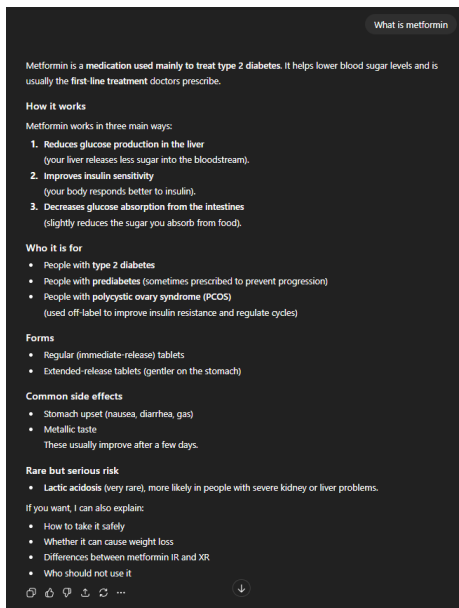
Department: Computer Engineering

Part 1: Prompt Comparison Analysis

The chosen domain we are going to use in this part is Healthcare. In the first prompt ChatGPT is going to be given a vague prompt asking about a medicine. In the second prompt, by using role prompting technique, the same prompt will be given and results will be analyzed.

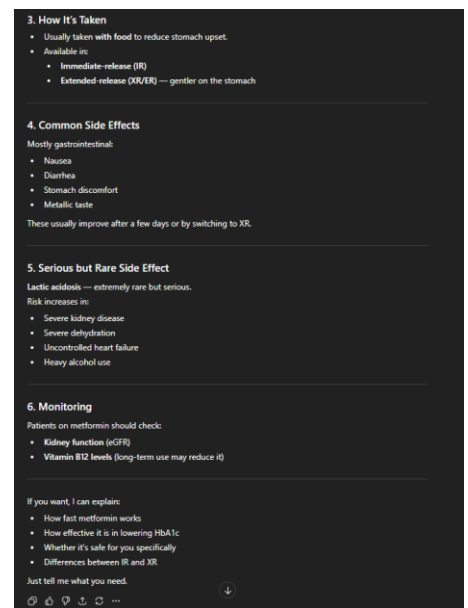
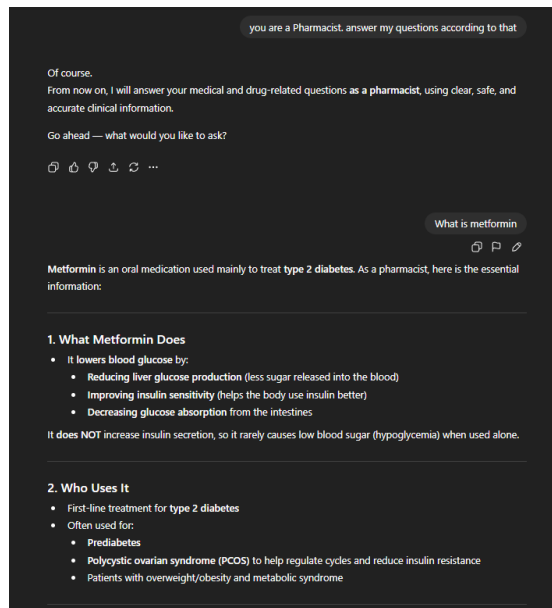
Version A Vague Prompt:

What is Metformin ?



Version B Improved Prompt (Role prompting):

What is Metformin ?



As we can see in the first prompt, ChatGPT gave us a simple and general explanation about the medicine which many people find enough for their needs. But in the second prompt, When the Pharmacist role is given to ChatGPT the response is more detailed, more complicated and including explanation about more sides off the medicine that aren't used in the daily life. Both of the prompts are useful depending on who is using the prompt and for what purpose. For

medical department student, using role prompting may be more useful compared to the casual prompt because they may need more detailed explanations and more information about all the good-bad or other sides of the medicine. In most cases the basic prompt is enough for people who just want a simple clarification about the medicine, the cause of using it or the advantages and disadvantages of it. In terms of the AI response, using techniques like this is going to give better answers because without using any of the techniques the AI is going to behave like a normal person answering a question, not like an expert but on the contrary when a technique is used, the AI is going to analyze the question as a person who has more knowledge in the concept given and give more clarified and specified answer to the topic.

Part 2: Practical Prompt Design

For this task, I chose CO-STAR Framework because it can give better results in terms of business. The task is to create a social media content calendar for one month and CO-STAR Framework combined with the system of ChatGPT can give us a good reliable output. CO-STAR Framework stands for many important sides of a framework and simply it is:

C – Customer:

Who is the target user or beneficiary?

*Which group has the problem?

*What do they need?

O – Opportunity:

What is the problem or gap you are solving?

*Why does it matter now?

*What pain point or inefficiency exists?

S – Solution:

What is your proposed idea or product?

*How does it solve the opportunity identified?

T – Team:

Who will execute the idea?

*What capabilities, skills, or resources does the team have?

A – Advantage:

What makes your solution unique or better?

*Your competitive edge or differentiator.

R – Results:

What outcomes or impact do you expect?

*Benefits, metrics, value created, or success indicators.

Create a social media content calendar for one month in english using CO-STAR Framework

Below is a one-month social media content calendar written entirely in English, structured using the CO-STAR Framework (Context – Objective – Style – Tone – Audience – Response).
You can adapt it for Instagram, TikTok, Facebook, or X.

One-Month Social Media Content Calendar (CO-STAR Framework)

WEEK 1

Post 1 — Motivational Carousel

Context: Start the month with uplifting, shareable content.

Objective: Boost engagement and save-rates.

Style: Short punchlines, clean layout.

Tone: Encouraging, warm.

Audience: Youth (18–30) seeking inspiration & direction.

Response: “Save this for later” comments, shares, positive reactions.

Content Idea:

“5 Mindset Shifts for a Better Week”

Slides include:

1. Progress > perfection
2. Small steps matter
3. Protect your energy
4. Choose learning over ego
5. Be patient with yourself



Post 2 — Reels (Behind-the-Scenes / Day in Life)

Context: Humanize the brand.

Objective: Increase authenticity & trust.

Style: Fast cuts, lo-fi sound, real moments.

Tone: Friendly, relatable.

Audience: Students & young professionals.

Response: Comments such as "I needed this!" and "Same vibe today."

Content Idea:

A short reel of your workspace, coffee moment, notebook planning, city street clip.

Text: "Building something great starts with small daily rituals."

Post 3 — Educational Static Post

Context: Provide value & position the account as a learning source.

Objective: Saveable educational content.

Style: Minimalist infographic.

Tone: Clear, instructive.

Audience: Learners, curious followers.

Response: Saves + shares.

Content Idea:

"3 Productivity Mistakes You Don't Notice You're Making."

WEEK 2

Post 1 — Inspirational Quote Reel

Context: Mid-month energy boost.

Objective: Reach new audiences through Reels.

Style: Cinematic clips + bold text overlay.

Tone: Emotional, heartfelt.

Audience: Young adults facing life transitions.

Response: Shares in stories.

Content Idea:

Quote: "Your future needs you more than your past does."

Post 2 — Topic Explainer Carousel

Context: Grow authority in your niche.

Objective: Teach a concept simply.

Style: Step-by-step slides.

Tone: Smart yet accessible.

Audience: Curious learners, self-improvement followers.

Response: Saves + discussion in comments.

Content Idea:

“What Is Emotional Discipline? (And Why It Matters)”

Post 3 — Poll / Question Story

Context: Increase interaction.

Objective: Learn about your audience.

Style: IG story poll, slider, Q&A.

Tone: Conversational.

Audience: Active followers.

Response: High story engagement.

Content Idea:

“What’s stressing you the most this week?” with poll options.

WEEK 3

Post 1 — Reels: Micro-Story

Context: Reels dominate reach.

Objective: Increase follow-rate.

Style: Storytelling + captions.

Tone: Inspirational.

Audience: Youth seeking relatable stories.

Response: Comments + shares.

Content Idea:

A 10–12 second reel narrating a short personal win or lesson learned.

Post 2 — Educational Carousel

Context: Provide deep value mid-month.

Objective: Establish thought leadership.

Style: Bold headers, digestible text.

Tone: Professional but friendly.

Audience: Students, creators, young professionals.

Response: Saves.

Content Idea:

“How to Think Clearly: 4 Evidence-Based Techniques.”

Post 3 — Community Highlight Post

Context: Build belonging and trust.

Objective: Strengthen community identity.

Style: Photo collage or testimonial graphic.

Tone: Appreciative.

Audience: Loyal followers.

Response: Tagging friends.

Content Idea:

Feature a follower comment, DM, or question + reply publicly.

WEEK 4

Post 1 — Monthly Reflection Reel

Context: End-of-month momentum.

Objective: Encourage reflection & saves.

Style: Soft music + text overlays.

Tone: Calm, introspective.

Audience: Followers seeking self-growth.

Response: Saves + shares in stories.

Content Idea:

“3 Questions to Ask Yourself Before the Month Ends.”

Post 2 — Tips / How-To Carousel

Context: Deliver tactical value.

Objective: Inform & educate.

Style: Step-by-step visuals.

Tone: Practical, concise.

Audience: Doers, planners, goal-oriented users.

Response: Saves + shares.

Content Idea:

"How to Plan the Next Month in 15 Minutes."

Post 3 — Announcement / CTA Post

Context: Prepare for next month's strategy.

Objective: Convert followers into active participants.

Style: Bold, branded.

Tone: Confident.

Audience: Whole community.

Response: Comments + sign-ups (if applicable).

Content Idea:

Ask followers what content they want next month + CTA to engage.

In this prompt we used "CO-STAR Framework" technique with the prompt "Create a social media content calendar for one month in English using CO-STAR Framework". The result we got is specified well to the task given but we didn't make some of the important parts of the request clear such as the calendars use purpose or justify a specific timeline, time zone or region of use for the AI to give use more proper and usable calendar for the plan we have.

Part 3: Critical Reflection

My understanding of prompt engineering has evolved over the first half of this course from seeing prompts as casual questions to recognizing that they are carefully designed instructions capable of guiding an AI system with astonishing precision. Early in my learning, I believed that prompting was basically just asking a model for what I wanted. With time, however, I learned that the structure, clarity, and level of detail inside a prompt can make the output quality differ completely. A major aha moment for me came when I realized how adding context and constraints such as defining the audience, specifying tone, or outlining the exact format significantly improved the results. This helped me shift from guessing what might work to intentionally designing prompts with purpose. Probably the most challenging concept so far was to understand how breaking down complex tasks into smaller steps rather than expecting the model to handle everything at once was better. I overcame it by practicing iterative prompting, rewriting my instructions in several rounds, and comparing the differences each adjustment made. One real-world scenario where I can apply this prompt engineering in my field is academic research support; many times, I have to summarize really long articles, organize references, improve clarity in language, or just generate structured outlines of conference papers. In this instance, I would be relying on frameworks such as CO-STAR or role prompting because they definitely help in setting clear expectations of what responses to expect and reduce vagueness or overly broad responses. For instance, assigning the AI the role of research assistant and specifying what the task is, what tone it should be written in, and adding other constraints usually yields much more accurate and usable results. However, I do foresee a number of challenges, among them the risk of hallucination, whereby the model provides incorrect information or fabricated sources with so much confidence. This I would address by incorporating explicit instructions such as do not make up citations or state uncertainty where unsure, and always cross-checking important points against reliable references. Secondly, there is the challenge of maintaining academic integrity, especially when rewriting or expanding on content using AI. In handling it responsibly, I would make sure that

the AI only helps with clarity and structure while ensuring that the original ideas and arguments remain mine. There are also critical limitations to remember: first, language models do not actually understand the meaning of content-they generate patterns based on data-so their answers have to be used carefully; second, because their knowledge is bound by their training cut-off, they may miss recent developments. Ethically, I also have to consider issues of transparency and fairness in the use of AI. One should not misrepresent entirely original work from AI but also avoid creating biased or harmful information by framing certain prompts in a way that creates such information. Assuring responsible use involves giving clear and safe instructions and remembering above all that AI is supposed to support, not replace, human judgment. For the second half of the course, I will be interested in learning more advanced prompting techniques, particularly multi-step workflows and evaluation prompts that help enhance accuracy and consistency. By the end of this course, I hope to close my skill gap in the design of prompts that can reliably produce high-quality and academically sound outputs with minimal revision.